

Ben Junghwan Lee

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EDUCATION

Bachelor of Fine Arts in Industrial Design — Brigham Young University

April 2013

PROFESSIONAL EXPERIENCE

Industrial Design Lead — **Axiom Space**

May 2020 – Present

Webster, TX

- **Multidisciplinary Leadership:** Direct end-to-end design and manufacturing alignment for spaceflight hardware and crew systems in close collaboration with space architects, soft goods specialists, and engineering teams.
- **Cross-Functional Mentorship:** Mentor junior designers and spearhead stakeholder design workshops to drive unified user experience (UX) goals across complex, high-stakes aerospace programs.
- **Rapid Prototyping & Simulation:** Validate advanced architecture concepts utilizing full-scale functional mockups, rapid tactile prototypes, and immersive AR/VR simulation environments.
- **Design Review Governance:** Lead formal internal design reviews and critical astronaut feedback sessions to mature flight-ready concepts through rigorous operational milestones.
- **Manufacturing Integration:** Partner with external vendors in weekly engineering reviews to preserve design intent while balancing structural, ergonomic, and production efficiencies.
- **Brand Strategy:** Establish overarching design brand guidelines and comprehensive Color, Material, and Finish (CMF) standards for future commercial habitat modules.
- **Cross-Departmental Consulting:** Deliver strategic UX and design consultancy across Crew Systems, Structural, ECLSS, Avionics, and Marketing departments.

Industrial Designer III — **Bigelow Aerospace**

June 2017 – March 2020

Las Vegas, NV

- **Habitat Systems Architecture:** Designed and prototyped Environmental Control and Life Support Systems (ECLSS) and Crew Systems for expandable space habitats under the NASA NextSTEP initiative, including crew quarters, lavatories, hygiene booths, exercise gear, and stowage systems.
- **Aerospace Compliance:** Developed high-reliability concept models meeting stringent standards for launch forces, microgravity safety, weight optimization, and human factors.
- **Astronaut Testing & Validation:** Partnered directly with NASA astronauts in ground testing to optimize equipment ergonomics, operational workflows, and microgravity usability.
- **High-Fidelity Delivery:** Fabricated 1:1 scale operational models, concept sketches, and detail-driven form/fit/function prototypes using advanced machinery (CNC mills, 3D printers, laser/vinyl cutters) across a vast matrix of aerospace and composite materials.
- **Executive Communication:** Presented complex design architectures to peers, executives, and leadership through advanced visualization and physical scale models.

Industrial Designer — **Owlet Baby Care**

January 2013 – February 2017

Lehi, UT

- **Iterative Prototyping & User Research:** Prototyped daily to accelerate design verification; planned and executed hundreds of in-home and remote user behavior tests to synthesize qualitative and quantitative data.
- **Global Manufacturing & Sourcing:** Sourced materials, evaluated domestic and international facilities, and established soft goods manufacturing assembly lines from scratch.
- **Quality & Engineering Alignment:** Collaborated with factory managers and engineers to resolve production roadblocks, resulting in design innovations that elevated product safety and reduced BOM costs.
- **Intellectual Property:** Named co-inventor on **two design patents** for consumer infant wearable devices.

Design Consultant — **Freelance**

January 2013 – Present

Remote / Various

- **End-to-End Client Strategy:** Guided diverse clients on design direction, scoping, timelines, and budgets from initial ideation to mass manufacturing.
- **Diverse Industry Execution:** Delivered specialized services—including rapid prototyping, logo design, front-end app development, circuit board design, and Kickstarter consultancy—across medical, consumer electronics, outdoor sports, and food sectors.